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1. INTRODUCTION TO NETWORKING

Here is the link to my completed module:

<https://academy.hackthebox.com/achievement/2386390/34>

A) INTRODUCTION

1.1 Networking Overview

In this section, I am able to get the general knowledge on networking from this module. Networking allows devices to communicate using various protocols, transmission media and topologies. I learnt about the flat network which are simple to deploy and function reliably providing minimal security. It talked of poor network design and a lack of subnetting awareness can cause loss of high-value assets. Overall, effective network segmentation reduces attack surfaces and prevents common exploits.

1.2 Network Types

In this module, I got a refresher on network types which determine how networks are connected and accessed. On the section, I got more details on the WAN, LAN, WLAN and the VPN; how they work and their capabilities.

B) NETWORKING STRUCTURE

1.3 Network Types

This section explained terminologies on Network Topology and the different kinds of them. Physical topology mainly focuses on how devices are physically connected using wired or wireless media while the logical topology describes how data flows across the network regardless of the physical layout. The Network Topology can be classified by their connections, nodes, structure and commonly grouped into 8 types which have unique characteristics affecting both performance and reliability. The knowledge from this helps in cybersecurity by anticipating failure points and implement appropriate security and monitoring controls.

1.4 Proxies

Proxy is basically defined as a mediator that sits in the middle of a connection can inspect and act on traffic. Proxies have specific roles depending on their type and use case. Forward proxies handle outgoing client requests and are commonly used to filter traffic, enforce security controls and control communication. On the other hand, reverse proxies filter incoming traffic to protect internal services, inspect requests and absorb attacks. The main functions of the proxies are; mediating, inspecting, visibility and controlling network to improve security.

C) NETWORKING WORKFLOW

1.5 Networking Models

On this section, I get to get an insight on some of the networking frameworks like OSI and TCP/IP models which show the transmission of data between hosts by breaking communication into layered processes that transform raw bits into usable information. Understanding the concept in reference to cybersecurity allows traffic to be assessed and well secured.

1.6 The OSI Models

In this section, I understand that the model was designed as a standardized reference framework to enable dependence in operation between different systems by breaking network communication into seven hierarchical layers. By running through all the layers on both ends of a connection, the model ensures reliable, efficient and secure communication while offering a clear way to assess how data flows through the network.

1.7 The TCP/IP Models

This module talks about the TCP/IP models being a layered reference framework built around the protocols that power modern networks and the internet having the TCP at the transport layer and the IP at the network layer. It enables applications to communicate across any network regardless of location. The TCP ensures reliable session management, data transfer and flow control while the IP handles logical addressing and routing.

D) ADDRESSING

1.8 Network Layer

This section describes the Layer 3 which controls the exchange of data packets, as these cannot be directly routed to the receiver and therefore have to be provided with routing nodes. At this layer, packets are not processed by higher-layer protocols on intermediate devices; instead, routers simply determine the next hop and forward the data accordingly. By managing packet forwarding within and between subnets, the network layer forms the backbone of reliable and scalable communication across complex networks.

1.9 IP Addresses

On this module, I get insight on network communication which relies on both MAC and IP addressing, where MAC addresses identify devices within a local network while IPv4 or IPv6 addresses enable communication across different networks and the Internet. IP addresses function like postal addresses by identifying the network and host portions needed to deliver data to the correct destination, while routers use this information to forward packets between subnets. Understanding binary representation, subnet masks, gateways, and broadcast behavior is essential for designing, troubleshooting, and securing networks effectively.

2.0 Subnetting

This is the process of dividing a larger IPv4 address range into smaller, logical networks that share the same network address, allowing administrators to secure and manage traffic. By applying a subnet mask, an IP address is split into a fixed network portion and a variable host portion, which determines the network address, broadcast address, and the range of usable host addresses within each subnet. Network addresses identify the subnet itself; broadcast addresses allow communication to all hosts in that subnet, and the remaining addresses are assigned to devices. Subnetting works entirely in powers of two, making it possible to calculate subnet sizes, host counts, and address ranges by adjusting CIDR notation and subnet masks.

Questions

1. Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27
255.255.255.224

The screenshot shows a web browser window with the URL `academy.hackthebox.com/module/34/section/306`. The page content includes a table with powers of two and their corresponding subnet masks, followed by explanatory text about IP address ranges. A question is presented: "Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27". The answer `255.255.255.224` is entered in the text field, and a "Submit" button is visible.

	6	4	2 ²	256/2/2/2/2/2
7	2	2 ¹	256/2/2/2/2/2	

By remembering the powers of two up to eight, it can become an instant calculation. However, if forgotten, it may be quicker to remember to divide 256 in half the number of times of the remainder.

The tricky part of this is getting the actual IP Address range because 0 is a number and not null in networking. So in our /25 with 128 IP Addresses, the first range is 192.168.1.0-127. The first address is the network, and the last is the broadcast address, which means the usable IP Space would become 192.168.1.1-126. If our IP Address fell above 128, then the usable ip space would be 192.168.1.129-254 (128 is the network and 255 is the broadcast).

☐ Enable step-by-step solutions for all questions

Questions

Answer the question(s) below to complete this Section and earn cubes!

+2 📦 Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

255.255.255.224

Submit

2. Submit the broadcast address of the following CIDR: 10.200.20.0/27
10.200.20.31

The screenshot shows the same web browser window as the previous one, but with two questions. The first question is identical to the one in the first screenshot. The second question asks for the broadcast address of the CIDR 10.200.20.0/27, with the answer `10.200.20.31` entered in the text field.

The tricky part of this is getting the actual IP Address range because 0 is a number and not null in networking. So in our /25 with 128 IP Addresses, the first range is 192.168.1.0-127. The first address is the network, and the last is the broadcast address, which means the usable IP Space would become 192.168.1.1-126. If our IP Address fell above 128, then the usable ip space would be 192.168.1.129-254 (128 is the network and 255 is the broadcast).

☐ Enable step-by-step solutions for all questions

Questions

Answer the question(s) below to complete this Section and earn cubes!

+2 📦 Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

255.255.255.224

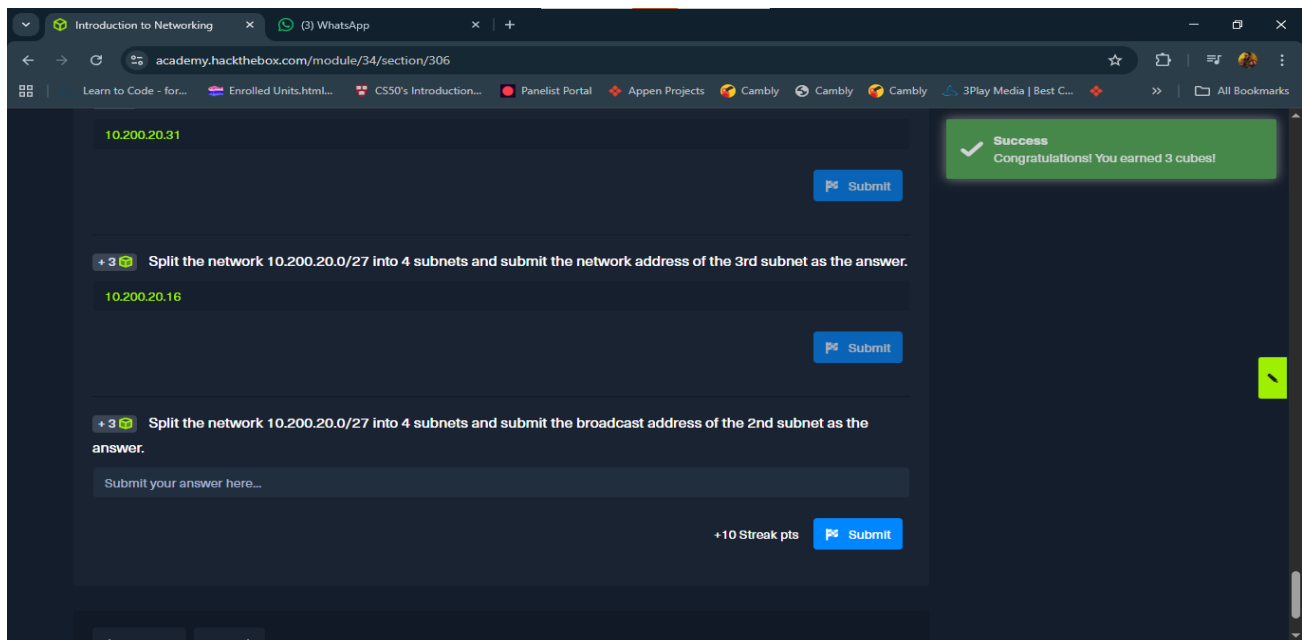
Submit

+2 📦 Submit the broadcast address of the following CIDR: 10.200.20.0/27

10.200.20.31

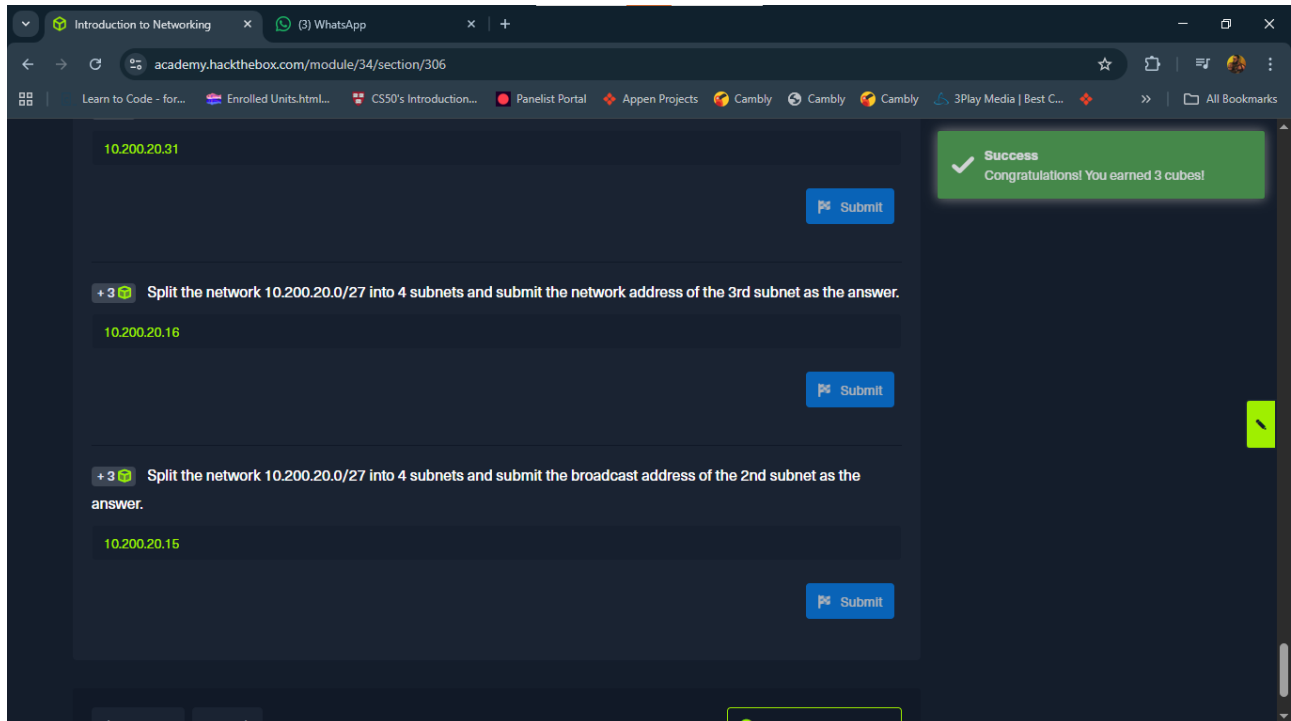
Submit

3. Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.
10.200.20.16



4. Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

10.200.20.15



2.1 MAC Addresses

This section describes the importance of MAC addresses which are unique 48-bit physical addresses assigned to network interfaces to enable communication at the data link layer. They are represented in hexadecimal format and consist of two main parts: the Organizationally Unique Identifier (OUI), which identifies the manufacturer, and the Network Interface Controller (NIC) portion, which uniquely identifies the device. MAC addresses are essential in local network communication, as devices use the Address Resolution Protocol (ARP) to map IP addresses to MAC addresses for packet delivery within the same subnet or through a router when communicating across subnets. Despite their importance, MAC addresses can be manipulated through techniques such as spoofing and flooding, which attackers may use to gain unauthorized access or disrupt network operations.

2.2 IPv6 Addresses

Got to learn about the advancement of IPv4 which brought in IPv6 which was developed to address the limitations of IPv4, particularly the shortage of available addresses. IPv6 uses 128-bit addresses, providing a vastly larger address space and enabling end-to-end communication without the need for Network Address Translation (NAT). IPv6 supports features such as automatic address configuration (SLAAC), faster routing, multiple addresses per interface, and built-in support for IPsec to enhance security. These improvements make IPv6 more scalable, efficient, and suitable for the continued growth of the Internet.

E)PROTOCOLS & TERMINOLOGY

2.3 Networking Key Terminology

This section gives an insight on key terminologies and common network protocols which is essential for building a strong technical foundation. Protocols such as HTTP, FTP and SMB enable communication, file sharing, and data transfer across networks, while security protocols like WPA, VPN and PGP help protect systems and data from unauthorized access. These concepts provide a critical foundation for further learning and specialization in areas such as networking, cybersecurity, and system administration.

2.4 Common Protocols

Internet protocols are standardized rules defined in RFCs that enable devices on a network to communicate reliably and consistently over wired or wireless connections. Understanding these protocols and their functions is essential for effective network management, troubleshooting, and cybersecurity.

2.5 Wireless Networks

This section describes Wireless Network which are communication systems that use radio frequency signals to connect devices such as laptops, smartphones, and tablets without physical cables, enabling access to shared resources and the Internet through wireless access points (WAPs). Wireless security is maintained through encryption, access control, and firewalls, with modern standards. The reliability and performance of wireless communication depend on factors such as signal strength, environmental obstacles and interference.

2.6 Virtual Private Networks

On this module, I learnt about the VPN and what it entails. A Virtual Private Network (VPN) is a technology that enables secure and encrypted communication between a remote device and a private network over the public internet, allowing authorized users such as administrators and employees to access internal resources safely from different locations. VPNs enhance security by encrypting data, authenticating users, and creating secure tunnels that protect communication from interception, while also providing cost-effective remote access and enabling organizations to connect branch offices into a unified network.

2.7 Vendor Specific Information

Here, I got to understand the concept behind Cisco IOS which is the operating system is used in Cisco networking devices such as routers and switches, providing essential features for managing and operating modern networks, including support for IPv6, Quality of Service (QoS), virtualization technologies, and security mechanisms like encryption and authentication. Cisco Discovery Protocol (CDP) helps administrators discover directly connected Cisco devices and understand network topology, while the Spanning Tree Protocol (STP) prevents network loops by ensuring a loop-free topology, both of which play important roles in network management, monitoring, and troubleshooting.

F) CONNECTION ESTABLISHMENT

2.8 Key Exchange Mechanisms

On this section, I was able to get information on Key Exchange Mechanisms. Key exchange methods are fundamental in cryptography because they allow two parties to securely establish a shared secret key over an insecure network, which is then used to encrypt and protect communication. Internet Key Exchange (IKE) builds on these cryptographic techniques to establish secure sessions in VPNs by negotiating keys and security parameters, operating in either main mode, which offers stronger security but slower performance, or aggressive mode, which is faster but provides less identity protection.

2.9 Authentication Protocols

Authentication protocols are critical in networking because they securely verify the identities of users, devices, and other entities, ensuring that only authorized parties can access sensitive resources. The choice of authentication protocol depends on the network environment, type of access, and required security level, with more robust protocols increasingly replacing older, vulnerable methods.

3.0 TCP/UDP Connections

Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) are key transport-layer protocols for transmitting data over the Internet, with TCP providing reliable, connection-oriented delivery for critical data like web pages and emails, and UDP offering faster, connectionless delivery for real-time applications. Security concerns include blind spoofing, where attackers manipulate IP headers to intercept or disrupt communications without seeing responses.

3.1 Cryptography

On this section, I learnt that encryption is used on the internet to transmit data, such as payment information, e-mails, or personal data, confidentially and protected against manipulation. In principle, we can distinguish between symmetric and asymmetric encryption techniques. Asymmetric methods have only been known for a few decades. Nevertheless, they are the most frequently used methods in digital communication. Block ciphers use modes such as ECB, CBC, CFB, OFB, CTR, and GCM to process data, each suited for different scenarios, including real-time communication, disk encryption, and applications requiring both confidentiality and integrity.

G) SUMMARY AND CONCLUSION

In conclusion, networking enables devices to communicate and share data efficiently, relying on structured workflows, addressing schemes, and protocols to ensure reliable and secure transmission. Security is maintained through encryption methods, key exchange algorithms which protect data integrity, confidentiality, and user identity. Modern networking combines architecture, protocols, encryption, and authentication to provide efficient, reliable, and secure communication, supporting normal day-to-day and enterprise-level applications.

